Artem Semidetnov

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Education

BSc	**Saint-Petersburg State University**, Mathematics	2021 – 2025
BSc	**Neapolis University Of Pafos**, Applied Computer Science JetBrains Program	2022 – 2024
MSc	**Geneva University**, Mathematics	2025 – 2027

Research Interests

- Categorical homotopy theory; simplicial methods and operads; homotopy type theory and computer-formalized approaches; geometric and combinatorial group theory.

Awards and Scholarships

Excellence Master Fellowship at Geneva University

Scholarship by "Rodnye Goroda" (a social investment program of PJSC "Gazprom neft"), 2023-2024

JetBrains Scholarship in Neapolis University

Travel grant of Chebyshev laboratory for Winter student school in mathematics and CS of HSE and MCS, Moscow

Publications

Operadic double dual (preprint in preparation)
Arshak Aivazian, Emmanuel Farjoun, Vasily Ionin, ***Artem Semidetnov***

The operad associated to a crossed simplicial group — Dec 2025
Artem Semidetnov · doi:10.48550/arXiv.2512.14498

On the geometry of free nilpotent groups — May 2021
Artem Semidetnov, Ruslan Magdiev · doi:10.48550/arXiv.2106.00095

AI-Assisted Mathematical Research

- An explicit non-elementary matrix over a bivariate Laurent polynomial ring — Vasily Ionin, ***Artem Semidetnov*** (Sept 2026).
- On Some More Problems from the Kourovka Notebook — Vasily Ionin, ***Artem Semidetnov*** (Aug 2026).
- The stable commutator length of a relator is not a one-relator group invariant — ***Artem Semidetnov*** (Aug 2026).

Other Mathematical Work

On Simplicial Resolutions for Link Groups
Artem Semidetnov

Work Experience

JetBrains, Research Intern in HoTT and Dependent Types Lab — Remote, Nov 2025 – Apr 2026

- Developed AI features for the Arend language including MCP servers for typechecking, IDE integration. Contributed to the Arend plugin in Intellij IDEA.
- Developed an experimental LLM coding agent for writing Arend code using framework Koog.
- Fine-tuned models for Arend code completion.

JetBrains, Research Intern in HoTT and Dependent Types Lab — Pafos, Cyprus, June 2024 – Sept 2024

- I was developing the official library in the Arend language. I formalized different results in algebra and homotopy type theory, including following.
- Formalized Eckmann-Hilton argument, Eilenberg-Maclane spaces, Homologies of types.
- Formalized automorphisms of groups, Schur's Lemma, Maschke's Lemma, Group actions characterizations. (Some of these results are in the 1.10 release).

IPONWEB (acquired by Criteo), Machine Learning Intern — Pafos, Cyprus, May 2023 – Sept 2023

- Criteo has a ML tool that analyzes sites and produces word2vec representations. Reverse-engineered the behaviour of this tool and analyzed its possible applications.

Teaching Experience

Sirius educational center, Teaching assistant on the course "Groups of intermediate growth" — Sochi, Russia, Apr 2024 – May 2024

Laboratory for continuous mathematical education, mathematics teacher for gifted students — Saint-Petersburg, Russia, 2021 – 2023

Mathematics and Computer Science faculty program for prospective students, Teaching assistant on the course Braid and knot theory — Saint-Petersburg, Russia, June 2023

Selected Talks

On the geometry of free nilpotent groups — Novosibirsk, Russia, 2021
- Siberian summer conference

"On the Poisson boundary of lamplighter groups" — Saint-Petersburg, Russia
- St. Petersburg Seminar on Representation Theory and Dynamical Systems

"Twisting numbers on braid and Thompson's groups" — N.Novgorod, Russia, 2024
- Topological Methods in Dynamics and Related Topics VII

"Plus construction and simplicial methods" — Pushkin, Russia, Aug 2025
- Conference "Algebraic Topology, geometry, combinatorics, and data analysis"

"The operad associated to a crossed simplicial group" — Copenhagen, Denmark, July 2026
- Young Topologists Meeting 2026

Schools and Workshops

Teaching Higher Category Theory with Computers — ICERM, Brown University, Providence, RI, Aug 2026
- Topical workshop on directed HoTT and computer-formalized higher category theory, organized by Emily Riehl, Nima Rasekh, Jonathan Weinberger. Participant (travel and lodging funded).

Miscellaneous

- Finalist of 2020 Intel ISEF
- Intel ISEF alumni
- Winner of 2019 Baltic SEF, PDMI special prize in 2019 Baltic SEF
- 3rd team place in 2019 International Tournament of Young Mathematicians in Barcelona, Spain
- Organizer of the Euler International Mathematical Institute's functional analysis seminar
- English level C1 (IELTS 8.0/9, taken in 2020, 2024)
- Invited judge in Saint-Petersburg Tournament of Young Mathematicians (since 2021)
- Created mathematical problem for 2024 International Tournament of Young Mathematicians 10th in here

- Invited judge in International Tournament of Young Mathematicians 2024, 2025
- A member of Scientific Organization Committee in ETEAM 2025